

Olympic Medal Analysis Report

Objective

To analyze Olympic medal data across years, countries, and disciplines using data visualization and Python-based data science techniques.

Dataset Overview

The dataset includes City, Year, Sport, Discipline, Event, Athlete, Gender, Country Code, Country, and Medal type. Encoding detected: ISO-8859-1.

Data Preprocessing

Used chardet to detect file encoding and pandas to load the data. Example:

```
with open("olympic medal.csv", "rb") as f:
```

```
    result = chardet.detect(f.read())
```

```
data = pd.read_csv("olympic medal.csv", encoding=result["encoding"])
```

Sample Data

Example row:

City: Montreal | Year: 1976 | Sport: Aquatics | Discipline: Diving | Event: 3m springboard | Athlete: KÖHLER, Christa | Gender: Women | Country: GDR

Visualizations

- Medal counts by country
- Distribution by gender
- Trends over years
- Top disciplines by medal count

Key Insights

- USA and Soviet Union top in medal counts

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- Male athletes dominate overall numbers
- Rising participation from women over time
- Country-specific strengths in disciplines